

- 1 In the light-dependent stage of photosynthesis, electrons and hydrogen ions are accepted by the coenzyme NADP, which becomes reduced.

DCPIP is a dye that can act as an electron and hydrogen ion acceptor. The dye is blue when oxidised and colourless when reduced. In laboratory experiments, DCPIP can be used to follow the progress of the light-dependent stage because it can replace NADP as the acceptor molecule for electrons and hydrogen ions, as shown in Fig. 1.1.

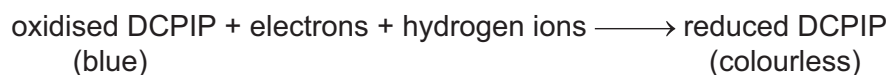


Fig. 1.1

The effects of various factors on the light-dependent stage of photosynthesis can be investigated by using suspensions of isolated chloroplasts (chloroplast suspensions) and DCPIP.

A student used DCPIP to investigate the effect of temperature on the rate of the light-dependent stage of photosynthesis in spinach, *Spinacia oleracea*.

The student prepared a leaf extract to make a stock chloroplast suspension and then carried out a preliminary experiment to determine a suitable concentration of chloroplast suspension to use in the investigation.

To carry out the preliminary experiment, the student followed a set of instructions, steps 1 to 11.

- 1 Cut spinach leaves into small pieces and place these pieces in a blender containing ice-cold 10% sucrose solution buffered at pH 7.0.
- 2 Turn on the blender for 15 seconds and then filter the extract to remove all the small pieces of leaf.
- 3 Place the leaf extract in a centrifuge and spin at low speed.
- 4 Pour off the supernatant that contains the chloroplasts. Keep this **stock chloroplast suspension** ice cold and in the dark.
- 5 Prepare 5 different concentrations of the chloroplast suspension using 10% sucrose solution. The percentage concentrations are 10%, 20%, 30%, 40% and 50% of the stock chloroplast suspension.
- 6 Wrap 5 flat-bottomed tubes in black plastic film to prevent light entering.
- 7 Put 10 cm³ of each concentration of stock chloroplast suspension into a flat-bottomed tube, and add 1 cm³ of DCPIP solution to each tube. The chloroplast suspension is now blue-green in colour.

- 8 Place 1 of the tubes beneath a light source as shown in Fig. 1.2.

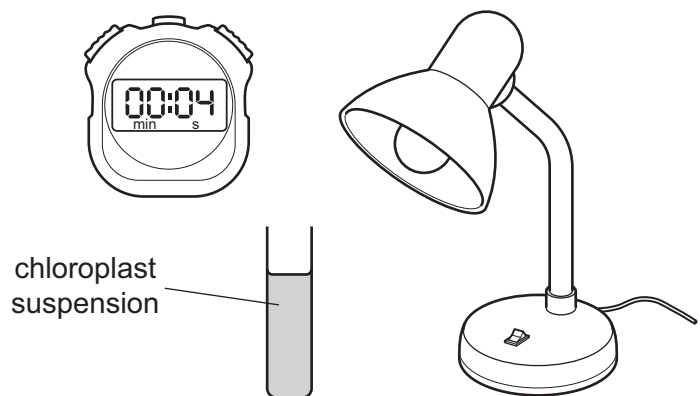


Fig. 1.2

- 9 Start a timer. Remove the black plastic film from the tube. Record the time taken for the DCPIP to decolourise so that the chloroplast suspension is green.
- 10 Calculate the rate of the light-dependent stage of photosynthesis by using the formula:

$$\text{rate} = \frac{1000}{t}$$

t = time taken in seconds for the chloroplast suspension to reach a green colour when all the DCPIP is decolourised.

- 11 Repeat step 8 to step 10 for the other tubes.
- (a) Suggest **two** suitable control experiments that the student should carry out as part of the preliminary experiment.

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(b) The results of the preliminary experiment are shown in Table 1.1.

Table 1.1

percentage concentration of chloroplast suspension	time taken for DCPIP to decolourise /s	rate of light-dependent stage of photosynthesis /s ⁻¹
10	351	
20	59	
30	21	
40	10	
50	5	

- (i) Complete Table 1.1 by calculating the rate of the light-dependent stage of photosynthesis for each concentration of chloroplast suspension.

Give your answers to **one** decimal place.

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- (ii) Plot a line graph of the data in Table 1.1 on the grid in Fig. 1.3 to show the effect of changing the concentration of the chloroplast suspension on the rate of the light-dependent stage of photosynthesis.

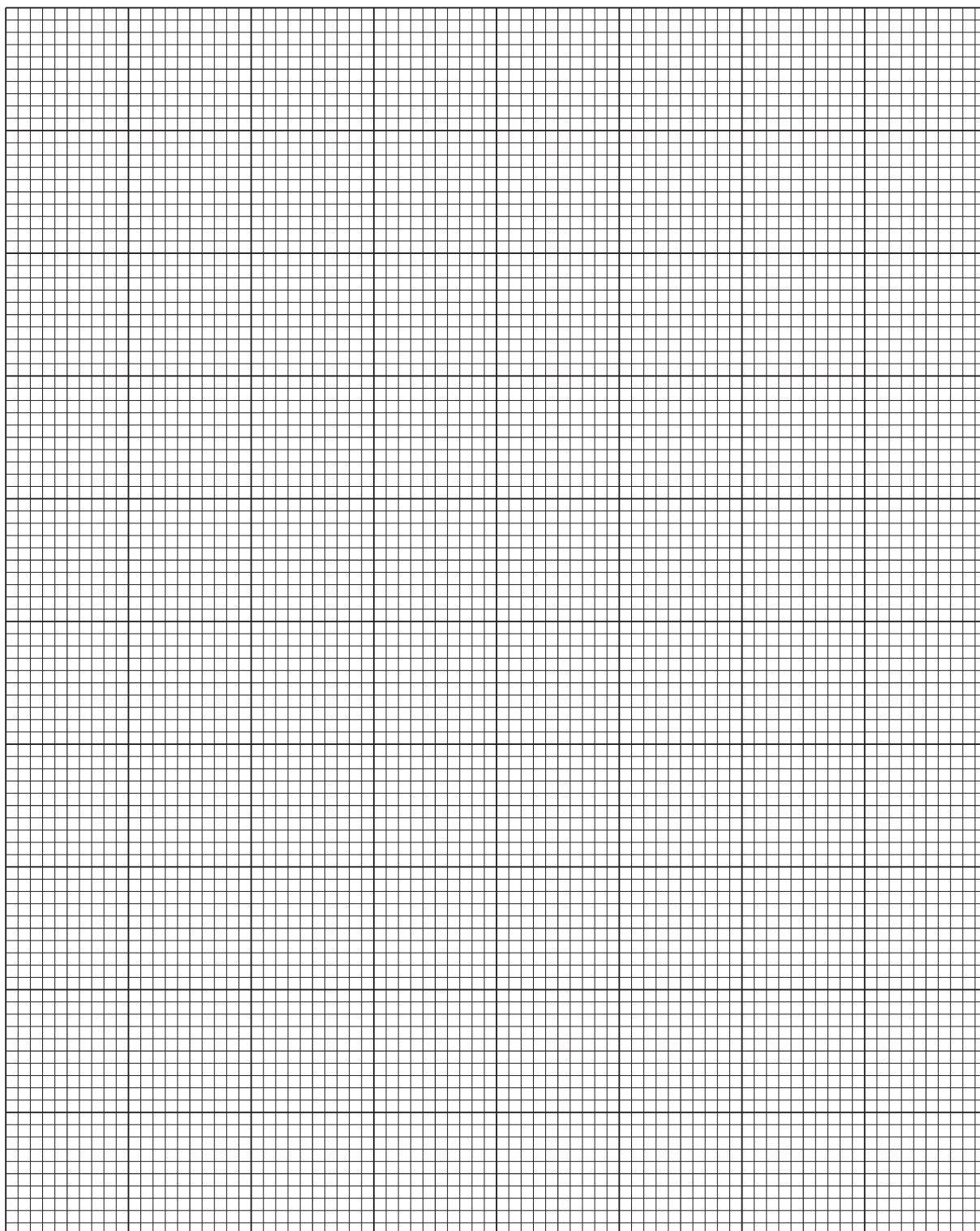


Fig. 1.3

(c) The student initially trialled the experiment using ice-cold distilled water instead of ice-cold 10% sucrose to make the stock chloroplast suspension in step 1. The DCPIP did **not** decolourise.

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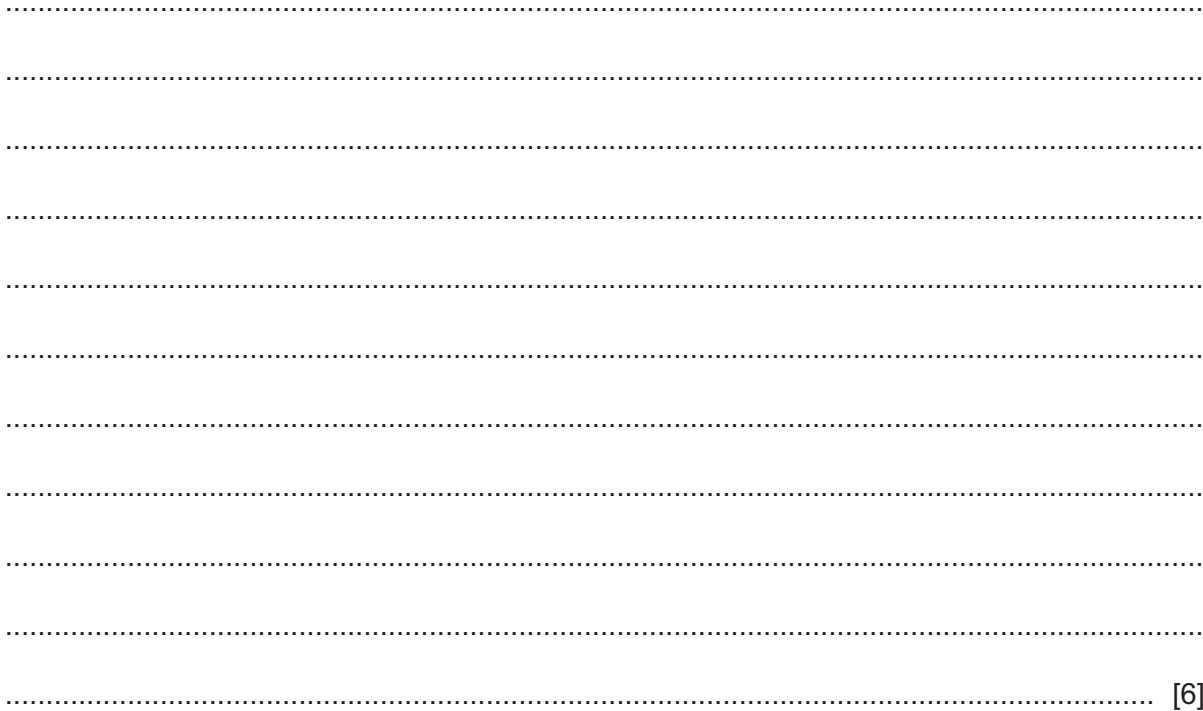
(d) The preliminary experiment was carried out at 17 °C.

To investigate the effect of temperature on the rate of the light-dependent stage of photosynthesis, the student added some details to the instructions in steps 7, 8 and 9 to make sure that accurate results were recorded.

Describe an investigation that the student could follow to determine the effect of temperature on the rate of the light-dependent stage of photosynthesis.

- Use the results shown in Table 1.1 to decide on a suitable chloroplast suspension for the investigation.
- Include details of how you would take accurate results.
- Do **not** include a risk assessment.

This image shows a blank sheet of white paper with ten sets of horizontal dashed lines, typical of primary-ruled notebook paper. Each set consists of three parallel dashed lines, providing a guide for letter height and placement. The lines are evenly spaced across the page.



[Total: 15]